Article

An Axiomatic Model IV: Axiomatic Identification of the Koide Parameters and the Scale Tier - the Amplitude 2 and the Phase Angle 2/9

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Abstract: Four axioms and one proposition define a discrete computational system. The present paper is the fourth of the Axiomatic Model series and proceeds on the same axiom set as Papers I, II, and III. The complete axiomatic system comprises 15 axioms [16]; the present paper additionally uses the number registry (complete descriptive degrees of freedom), the two reading rules, and the separation of system time and domain time. The present paper treats the Koide formula $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$. The $\mathbb{Z}_3$-symmetric parametrization of this relation was introduced by Koide, the parameter values $k_L = 1$ (amplitude $r^2 = 2$) and $\delta_L = 2/9$ (phase angle) were observed by Brannen and Rosen, and the formalism was organized by Zenczykowski. The present paper presents no new values. The present paper makes two claims. First, the two parameter values are identified with registered numbers of the axiomatic system: the amplitude $r^2 = 2$ is the two reversible axes, the phase angle $\theta = 2/9$ is the dimensionless ratio of the reversible-axis 2 to the structural complete descriptive degrees of freedom 9, and since reversible axes draw no arc, no π conversion can enter. Second, and as the central result of the present paper, the observed fact that the Koide relation holds only at the pole masses and breaks at high energy scales is shown to be a prediction of the axiomatic system: mass is the juim that CAS Swap records into DATA, hence a domain-time-tier quantity, and the domain-time tier includes thresholds, corresponding to the pole-mass definition. As a candidate for the departure size we present $Q(\mu) - 2/3 = \alpha(\mu)/(2\pi)$ (threshold-free scheme, Section 4.6). Since Paper I assigned the coupling constant $\sin^2\theta_W$ to the system-time tier and thereby explained its agreement with the threshold-free scheme, the present paper reinforces the opposite end of the same rule. The $\mathbb{Z}_3$ parametrization itself is the discrete Fourier transform of three real numbers, holds identically for any three positive numbers, and carries no physical input. The registered-number identifications carry postdiction status. The measured Q, propagating the tau-mass uncertainty, is 0.43σ from $2/3$ and hence indistinguishable from exactly $2/3$; the deviation coefficient is indiscriminable in both value and sign at current precision, and current data prefer the no-deviation hypothesis.

Keywords: Koide formula; charged lepton masses; Compare-And-Swap; pole mass; running mass; Z3 parametrization; neutrino masses; tau mass; computational primitives in physics

1. Introduction

The masses of the three charged leptons carry one celebrated relation, discovered by Yoshio Koide in 1981 [1,2]:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \tag{1}$$

With PDG 2024 values [10], $Q = 0.666664463 \pm 0.000005$, which is 0.43σ from $2/3$. That is, the measurement is indistinguishable from exactly $2/3$. The dominant error term is the tau-mass uncertainty ± 0.09 MeV.

The relation can be rewritten in a $\mathbb{Z}_3$-symmetric form [1,5]:

$$\sqrt{m_k} = \mu\left(1 + r\cos\left(\theta + \frac{2\pi k}{3}\right)\right), \qquad k = 0, 1, 2. \tag{2}$$

The measurements single out $r^2 = 2$ and $\theta = 2/9$ [3–5]. Several attempts at a theoretical ground exist: Foot's geometric interpretation [7], Sumino's family gauge symmetry effective theory [8], Koide's own yukawaon model [2], and the weak-basis interpretation of Gerard, Goffinet, and Herquet [6]. That is, it is not that no ground has been proposed; there is no agreed ground. And the relation leaves three questions.

- **(Q1) Amplitude**: why $r^2 = 2$?
- **(Q2) Phase angle**: why $\theta = 2/9$, and why does no π enter although it is an angle?
- **(Q3) Scale**: why does the relation hold only at the pole masses and break at high energy scales? (Observed by Zenczykowski [5])

The present paper answers the three questions. Q1 and Q2 are answered by identifying the two parameter values with registered numbers of the axiomatic system (Sections 5 and 6, Class III), and Q3 is answered by the scale-tier theorem derived from the axioms (Section 4). The answer to Q3 is the central result of the present paper.

Position of the paper. Paper I showed that the irreversible/reversible classification of cost forces the quadratic-form signature $(5,2)$ and reached the fine-structure constant [13]. Paper II forced the bit reading of the Wheeler-DeWitt equation from the same $(5,2)$ [14]. Paper III derived the three structural constants of black hole thermodynamics [15]. The present paper treats the fourth physical domain of the same axiom set, the generation structure, and in particular reinforces the opposite end of the scheme-correspondence rule that Paper I established on the system-time tier. Table 1 separates what the present paper presents from what it does not.

Table 1. Prior work and the position of the present paper.

Item	Prior work	Position of the present paper
Koide relation $Q = 2/3$	discovered by Koide, 1981 [1,2]	adopted as is; not new
$\mathbb{Z}_3$-symmetric parametrization	introduced by Koide [1], organized by Zenczykowski [5]	adopted as is; not new
Amplitude $k_L = 1$ (i.e., $r^2 = 2$)	confirmed empirically [3,5]	identified with a registered number (Section 5, Class III)
Phase angle $\delta_L = 2/9$	observed by Brannen and Rosen [3,4]	identified with a registered number (Section 6, Class III)
Interpretation of $k_L = 1$	weak-basis interpretation of Gerard, Goffinet, Herquet [6]; adopted by Zenczykowski [5]	an alternative identification; superiority not adjudicated (Section 7)
The 120-degree arrangement	introduced as a formalism [1,5]	the definition of the discrete Fourier transform; not a derivation target (Section 3)
Holding only at the low-energy scale	observed by Zenczykowski [5]	predicted from the axioms (Section 4, center of the present paper)

Organization of the paper. Section 2 presents the design principles, the terminology guide, the 4 axioms + 1 proposition, methodological transparency, and the exhaustive item list. Section 3 fixes the status of the $\mathbb{Z}_3$ arrangement. Section 4 derives the main result, the scale-tier theorem. Sections 5 and 6 identify the amplitude and the phase angle. Section 7 organizes the relation to prior interpretations. Section 8 presents the auxiliary results (the contraction overlap ratio rule, the neutrino sector, the Koide deviation). Section 9 discusses. Section 10 concludes.

Status classification. As in the earlier papers, the epistemic status of each result is classified in four classes: *I. Axiom-derived* (formal proof or forward chain completed), *II. Axiom-determined* (the axioms force a structure and mathematics fixes it automatically), *III. Axiom-described* (derivation path and interpretation are in the text; independent formal proof is subsequent), *IV. Out of scope* (not included in the scope of the present paper). The present paper assigns most items to III.

2. Axiomatic System

2.1. Design Principles

This section summarizes what the axiomatic system models and why it takes this form. The detailed design principles and physical motivation are in Paper I Section 2 [13]; this section records only the minimum needed to read the present paper self-containedly.

The axiomatic system consists of three parts.

1. *Structure*: four orthogonal axes (time, space, observer, superposition) are divided into two brackets, the classical bracket (DATA) and the quantum bracket (OPERATOR).
2. *Operator*: the single indivisible operation CAS (conditional state transition) proceeds in three steps, Read, Compare, and Swap. CAS operates on the OPERATOR side and records its result into the DATA bracket.
3. *Cost*: crossing an orthogonal boundary + in the prescribed order incurs cost +1; without order the cost is 0.

The cost rule classifies each axis as irreversible (cost > 0) or reversible (cost $= 0$), and Paper I proved that this classification forces the signature $(5, 2)$ [13]. What the present paper uses are the two reversible axes of that classification, the structural complete descriptive degrees of freedom 9, and the two time tiers of Axiom 1. The arrangement of the three parts is shown in Figure 1.

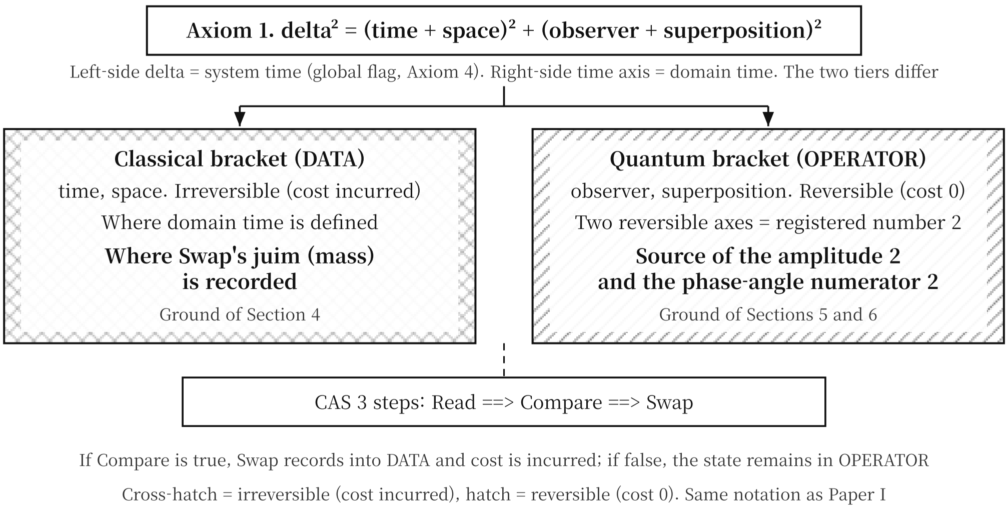

Figure 1. The structure of the Banya equation - the two brackets (classical DATA, quantum OPERATOR), the four domain axes, the CAS 3 steps, and the global δ. The two sites used by the present paper are marked: the recording site of the juim (DATA, Section 4) and the two reversible axes (OPERATOR, Sections 5 and 6).

2.2. Terminology Guide

The axiomatic system of the present paper uses a discrete operational structure called "conditional state transition." The structure is formally identical to what is known in parallel computing as Compare-And-Swap (CAS), and the present paper translates it into the physical context as follows. Hereafter "CAS" abbreviates "conditional state transition."

Table 2. Physical translation of the operational terms.

Physics term (this paper)	Computer-science counterpart	Remark
conditional state transition	Compare-And-Swap (CAS)	the indivisible operation observe ==> compare ==> record
observe (Read)	Read	fetch the current state
compare (Compare)	Compare	check against the expected value
record (Swap)	Swap	change the state when the condition holds
juim	grip	one state occupancy confirmed by Swap and recorded in DATA; mass is the accumulation of juims
order-enforcing device	lock	the constraint guaranteeing transition order
indivisible operation	atomic operation	a single transition that cannot be interrupted

2.3. Axiom Number Correspondence

The axiom numbers of the present paper follow the same 4-axiom system as the earlier papers and differ from the numbering of the complete 15-axiom system. A correspondence table is provided for machine checking.

Table 3. Axiom number correspondence.

This paper	Complete system [16]	Content
Axiom 1	Axiom 1	Banya equation, 4-axis orthogonal norm
Axiom 2	Axiom 2	CAS is the unique operator
Axiom 3	Axiom 4	cost
Axiom 4	Axiom 15	δ is the global flag
Proposition 1	Axiom 4 proposition	how to read cost as norm

2.4. The 4 Explicit Axioms and 1 Proposition

Axiom 1 (Banya equation). *Every change δ of the universe is the norm of the four axes:*

$$\delta^2 = \underbrace{(\text{time} + \text{space})^2}_{\textit{classical bracket (DATA)}} + \underbrace{(\text{observer} + \text{superposition})^2}_{\textit{quantum bracket (OPERATOR)}}. \tag{3}$$

The brackets are separators grouping the four axes in pairs, and the two brackets are orthogonal. The left-hand δ is system time, the right-hand time axis is domain time, and the two tiers differ.

Axiom 2 (CAS is the unique operator). *Every change is the repetition of the single CAS operation. CAS has the three steps Read, Compare, and Swap; the three steps are mutually orthogonal and occupy one bit each. The firing order is enforced by the lock.*

Axiom 3 (Cost). *Crossing an orthogonal boundary $+$ along the order incurs cost 1; without order the cost is 0. Order creates irreversibility and discreteness at once. There are five irreversible axes and two reversible axes.*

Axiom 4 (δ is the global flag). *The left-hand δ of the Banya equation is a global state flag outside the finite state machine of the right-hand side.*

Proposition 1 (How to read cost as norm). *The same $+$ crossing is read in two ways: cost reading counts it as the integer 1, and norm reading measures it as an arc length. An irreversible crossing is a hemisphere π; a reversible crossing is a full sphere 2π. The CAS forward direction is irreversible, hence π. Reversible axes incur cost 0, draw no arc, and admit cost-reading integers only.*

2.5. Scope of the Present Paper

The complete system comprises 15 axioms [16] and treats many physical constants beyond the Koide relation. The present paper uses only the same *4 axioms and 1 proposition* as the earlier papers, and, as items derived from them, the number registry (Section 2.9), the two reading rules (Proposition 1), and the separation of system time and domain time (Axiom 1). The exhaustive list of items used is given in Section 2.8. This is a self-contained account on the minimal axiom set and maximizes verifiability.

2.6. Reader's Guide: Four Common Misconceptions

Four items in which the notation and terminology of this axiomatic system differ from standard physics are stated in advance.

1. *The four axes are not 4D spacetime*: the four axes (time, space, observer, superposition) differ from Minkowski 4D spacetime. Spacetime is the two axes time + space (the DATA bracket); observer + superposition are the two axes of the quantum domain (the OPERATOR bracket). The four axes together do not constitute spacetime.
2. *Proposition 1 fixes two layers at once*: one is the *measure of a crossing* (irreversible π, reversible 2π), the other is the *accounting of reversible axes* (no arc, integers only). The former sets the size of the stage; the latter sets the notation of values on that stage. The two layers must not be confused. Section 3 uses the former; Section 6 uses the latter.
3. *The 120-degree arrangement is not a derived result*: the $\mathbb{Z}_3$ parametrization is the discrete Fourier transform of three real numbers and holds identically for any three positive numbers. The present paper does not derive it and uses it only as a notational device (Section 3).
4. *The axioms carry no notion of energy scale*: what the axiomatic system has instead is the tier separation of system time (δ) and domain time (the time axis). By the correspondence rule established in Paper I, this separation assigns to each physical quantity the scale at which its relation holds (Section 4).

2.7. Methodological Transparency

1. *Exhaustive statement of the items used.* "4 axioms + 1 proposition" is the logical skeleton. The actual account uses, besides the 4 axioms + 1 proposition, seven axiom-structure consequences (E1-E7), for a total of 12 items (listed exhaustively in Table 4). These are declared inputs, not hidden ones; each item is a logical consequence of the axioms or a result proved in the earlier papers.
2. *The registered-number identifications carry postdiction status.* The parameter values $r^2 = 2$ and $\theta = 2/9$ were observed first in the measurements [3,4]; the identifications of the present paper came later. The two identifications therefore carry Class III and are not derivations. This temporal relation is maintained throughout the text.
3. *What performs the exclusions is the comparison with measurement.* The dimensionless-ratio argument is true of every dimensionless number and excludes no candidate (Section 6.2). What excludes the π-form candidates is the measured mass reconstruction (Section 6.3).
4. *Non-uniqueness of the identifications is stated.* The same numbers 2 and 9 are registered in both the structural and the cost brackets, and the source cards read them differently (Appendices B and D). The existence of this freedom is what pins the identifications at Class III.
5. *Attribution of the values is stated.* The observation of the parameter values is due to Brannen and Rosen [3,4]; the organization of the formalism is due to Zenczykowski [5]. The present paper presents no new values (Section 7.2).
6. *What is designed and what is not.* The Banya equation (Axiom 1) was designed to be interchangeable with existing physical equations. The nontrivial claim is that the structure of this equation singles out specific numbers and a specific scale behavior that agree with measurement. In the present paper that claim is the scale-tier theorem (Section 4) and the departure-size candidate (Section 4.6).

2.8. Exhaustive List of Items

All items actually used in the account of the present paper are listed on an equal footing. Table 4 is the exhaustive item table; it makes explicit that the "4 axioms + 1 proposition" heading is not the complete item list.

Table 4. Exhaustive list of the items used in the account of the present paper.

ID	Item	Remark	Used in
A1	Axiom 1 (Banya equation)	explicit axiom; the δ^2 equation	throughout
A2	Axiom 2 (CAS 3 steps)	explicit axiom; R-C-S	throughout
A3	Axiom 3 (cost $+1$)	explicit axiom; declares the $(5,2)$ partition	Sec. 5, 6
A4	Axiom 4 (δ flag)	explicit axiom	Sec. 4
P1	Proposition 1 (how to read cost as norm)	explicit proposition; irreversible π, reversible 2π, no arc on reversible axes	Sec. 3, 6
E1	full sphere of a reversible crossing	direct application of Proposition 1: reversible ==> 2π	Sec. 3.2
E2	cost accumulation partition $(5,2)$	proved in Paper I Theorem 1 [13]	Sec. 5, 6
E3	structural complete descriptive DOF 9	$9 = 7 + 2$; registered in the complete system [16]	Sec. 6
E4	no arc on reversible axes	Proposition 1: cost 0, hence cost-reading integers only	Sec. 6
E5	separation of system time and domain time	immediate consequence of Axiom 1; made concrete by Paper I as the scheme-correspondence rule [13]	Sec. 4
E6	Compare branching	Axiom 2: on true, Swap records into DATA; on false, the state remains in OPERATOR	Sec. 4, 8.2
E7	recording site of the juim	Axiom 1 + Axiom 2: the juim made by Swap is recorded into DATA	Sec. 4

2.9. The Number Registry

The descriptive degrees of freedom divide into those derived from structure and those derived from cost. Combinations among registered numbers use only shifts within the same bracket and $+$ only across brackets.

The layer of the derivation column. The derivation column of the tables below describes why each number is registered; it is a different layer from the combination rule. Only actual combinations among registered numbers follow the rule.

Table 5. Structural descriptive degrees of freedom.

Structural DOF	Derivation	Role
1	minimal unit	bit basis
2	two brackets	DATA, OPERATOR
3	CAS 3 steps	R, C, S
4	four domains	observer, superposition, time, space
7	$T(3) + 1$	CAS internal degrees of freedom
9	$7 + 2$	structural complete descriptive DOF
16	2^4	4-axis domain combinations
30	$7 \times 4 + 2$	number of access paths
128	all 7-bit states	count of $\delta = 1$ valid states
137	$T(16) + 1$	data type maximum

Table 6. Cost descriptive degrees of freedom.

Cost DOF	Derivation	Role
1	minimum for crossing +	cost basis
2	two order-free axes	observer, superposition; cost 0
3	$R + C + S$, $+1$ each	CAS internal cost
4	gripping 3 axes + time	ball value
5	five irreversible axes	the $(5, 2)$ partition
9	$13 - 4$	residual cost
13	8 reads + 5 writes	total cost

3. The Status of the $\mathbb{Z}_3$ Arrangement

3.1. Not a Derivation Target

The 120-degree arrangement is not a derivation target. The discrete Fourier transform of three real numbers reproduces the three exactly with one mean term and one amplitude-phase pair, and the 120-degree arrangement is the definition of that transform itself. That is, Eq. (2) holds identically for any three positive numbers. The prior literature records no ground because none is needed for an identity.

The following table demonstrates this. For arbitrary triples of positive numbers, substituting the r^2 back-solved from the parametrization into $Q = (3 + \frac{3}{2}r^2)/9$ reproduces the measured Q exactly.

Table 7. Demonstration of the identity of the $\mathbb{Z}_3$ parametrization - recomputation for arbitrary positive triples.

Triple	Measured Q	r^2 back-solved	$(3 + 1.5r^2)/9$	Difference
$(1, 2, 3)$	0.3490095	0.09406	0.3490095	5.6×10^{-17}
$(0.001, 5, 900)$	0.8691871	3.21512	0.8691871	0
$(7, 7, 100)$	0.4875336	0.92520	0.4875336	0

Therefore the following holds. $Q = 2/3$ and $r^2 = 2$ are fully equivalent (Class II, pure algebra), but this equivalence carries no physical input. The present paper does not present it as a result and uses it only as a notational device in the later sections.

3.2. The Size of the Stage Is 2π

The size of the stage, however, can be fixed. By Proposition 1, a reversible crossing is a full sphere 2π and an irreversible crossing is a hemisphere π. Since the amplitude $r^2 = 2$ is the two reversible axes (Section 5), the circle on which the amplitude is located is reversible and its circumference is 2π. Here lies the ground for the three-point equal spacing $2\pi/3 = 120$ degrees.

This must not be confused with the π of the CAS forward path: that one is irreversible, hence a hemisphere. The two values are measures of different objects, and Proposition 1 fixes both together.

4. Main Result: The Scale Tier - Why the Pole Mass

4.1. The Problem

Quoting Zenczykowski, the simple numbers 1 and 2/9 "work well at the low-energy scale and not at some high mass scale," and the k_L and δ_L extracted from the charged-lepton masses at the M_Z scale depart from their perfect values [5]. That is, the Koide relation holds only at the pole masses and breaks at the M_Z scale. The axiomatic system carries no notion of energy scale. Why, then, is a specific scale singled out? This is the question of this section.

4.2. The Rule Established by Paper I

Paper I faced the same problem with the coupling constant and established the following rule [13]. One CAS cycle is one tick of system time, and the 13 + crossings occur within this one tick. The energy scale we measure, by contrast, is a quantity defined on domain time after rendering into DATA (the classical bracket). The two time tiers are different granularities, and the axiomatic system operates on the system-time tier only. The MS scheme is a threshold-free definition that explicitly removes physical particle-mass thresholds, and is therefore formally the nearer one to the system-time tier; the on-shell definition includes physical particle masses as thresholds and is nearer to the domain-time side.

Paper I's $\sin^2\theta_W = 7/(2+9\pi)$ is a system-time-tier quantity, hence agreed with the threshold-free scheme; it indeed agrees with the MS-scheme experimental value (to 4.5×10^{-6} on the Paper I basis, 3.0×10^{-4} against the PDG 2024 global fit) [13].

4.3. Koide Is the Opposite End

Theorem 1 (Scale tier)**.** *The Koide relation is a relation among domain-time-tier quantities; it therefore holds in the definition that includes thresholds, namely at the pole masses, and breaks at high energy scales. Class III.*

Proof sketch. Mass is the juim made by CAS Swap (Axioms 2, 3). The juim is recorded into DATA (the classical bracket) when Compare returns true. DATA is where domain time is defined (Axiom 1). Hence mass is a domain-time-tier quantity.

By the correspondence established in Paper I, the domain-time tier corresponds to the definition that includes physical particle masses as thresholds. The pole mass is exactly that definition. Hence structural relations among masses must hold at the pole masses. □

4.4. The Two Papers Reinforce Both Ends of the Rule

Table 8. Both ends of the same rule - tier and validity scale of the two quantities.

Quantity	What it is	Which bracket	Which time tier	Validity scale	Checked
$\sin^2\theta_W$	coupling constant	OPERATOR	system time	threshold-free (MS)	Paper I [13]
Koide Q	**result of juims**	**DATA**	**domain time**	**threshold-included (pole)**	**this paper**

Paper I checked the system-time case and the present paper checks the domain-time case. That the schemes of the two cases are opposite is the discriminating power of the rule: were the rule empty, the two cases would have pointed to the same scheme. Figure 2 summarizes the two chains branching from the same axioms.

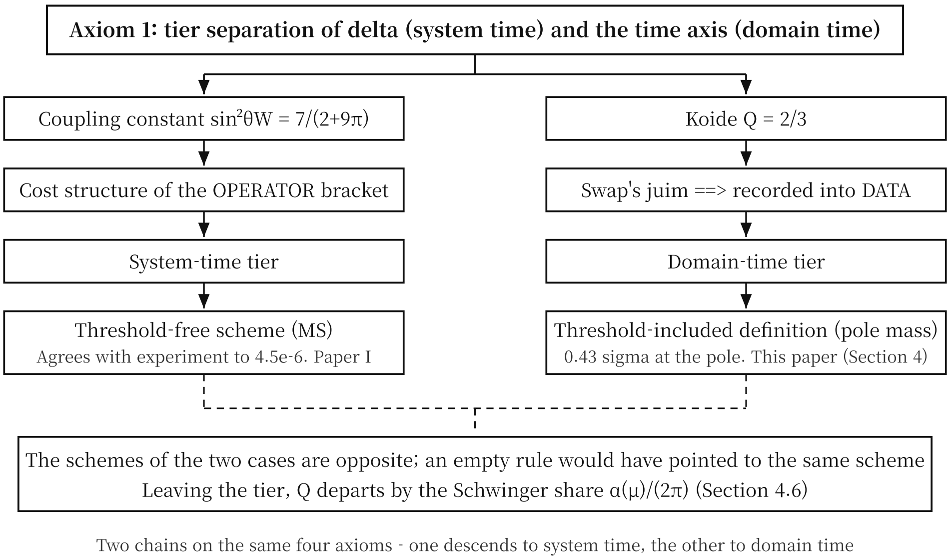

Figure 2. Two chains branching from the same axioms - the coupling constant descends to the system-time tier and holds in the threshold-free scheme (Paper I); the Koide relation descends to the domain-time tier and holds at the pole masses (this paper). The departure size at the tier boundary is Section 4.6.

4.5. Falsification Conditions

Table 9. Falsification conditions of Section 4.

Condition	Consequence
the Koide relation holds at the M_Z scale with the same precision as at the poles	this theorem is discarded
a case is found in which a coupling constant fits better in a threshold-included scheme	discarded together with Paper I's correspondence rule
the departure size disagrees systematically with the candidate $\alpha(\mu)/(2\pi)$	the size candidate of Section 4.6 is discarded; the direction prediction (this theorem) stands separately

4.6. A Candidate for the Departure Size (Class III)

We present a candidate formula for the size of the departure. Computing Q directly from the scale-dependent charged-lepton masses (the running-mass tables of Xing, Zhang, and Zhou [12]) gives $Q(M_Z) = 0.667925$, i.e., a departure of $+1.258 \times 10^{-3}$. This value agrees with the form

$$Q(\mu) - \frac{2}{3} = \frac{\alpha(\mu)}{2\pi}. \tag{4}$$

Structural reading. By Proposition 1 the stage of a reversible crossing is the full sphere 2π, and by Section 3.2 the circle on which the Koide amplitude is located is reversible with circumference 2π. Raising the scale is coupling-constant running; the moment mass leaves the pole (the domain-time tier) for the threshold-free scheme (the system-time side), Q departs by the share of one coupling spread over that reversible circle - the same form as the Schwinger term $a_e = \alpha/(2\pi)$ of the electron anomalous magnetic moment. Since the nearest integer to $1/\alpha$ at M_Z is $128 = 2^7$ (the $137 = 128 + 9$ of Paper I Section 8.1.2 [13]), the pure form in registered numbers only is $1/(2\pi \times 128) = 1/(256\pi) = 1.2434 \times 10^{-3}$.

Table 10. Scale-by-scale departure and required $1/\alpha$ - the all-scale check of the candidate formula.

Scale	$Q-2/3$	**Required** $1/\alpha$ = $1/(2\pi\, dQ)$	$1/\hat{\alpha}(\mu)$ **(literature anchors+1-loop)**	**Difference**
m_c (1.29 GeV)	$+1.180 \times 10^{-3}$	134.93	133.81	+0.8%
m_b (4.19 GeV)	$+1.200 \times 10^{-3}$	132.63	132.30	+0.2%
M_W	$+1.256 \times 10^{-3}$	126.76	128.11	−1.1%
M_Z	$+1.258 \times 10^{-3}$	126.51	127.930 (PDG)	−1.1%
m_t (163 GeV)	$+1.269 \times 10^{-3}$	125.38	127.11	−1.4%
1 TeV	$+1.258 \times 10^{-3}$	126.53	non-monotonic	outside range
pole	-8×10^{-6}	not applicable	departure effectively 0	-

Production of the $1/\hat{\alpha}$ column. The two PDG 2024 literature values $1/\hat{\alpha}(m_\tau) = 133.450$ and $1/\hat{\alpha}(M_Z) = 127.930$ are taken as anchors [10], and the intermediate scales are produced by one-loop running with active-fermion thresholds. The one-loop self-consistency between the two anchors is 0.01%. W-boson contributions are not accounted for; the two scales above M_Z are read within this limitation. The pole row of the table is on the XZZ input pole masses (tau 1776.82, the 2011 edition); updating the tau to the PDG 2024 value 1776.93 shifts the scale-wise dQ by about +0.5% - within the agreement digits of the table.

The required $1/\alpha$ descends monotonically from 134.93 to 125.38 along the scales, agreeing with QED running in direction, with the step size in the same order of magnitude. The sign of the difference flips at the electroweak threshold: +0.2 to +0.8% in the pure QED range (m_c, m_b), −1.1 to −1.4% in the electroweak range. In the 1 TeV row of the XZZ tables, $dQ = +1.258 \times 10^{-3}$ falls below the $+1.269 \times 10^{-3}$ of m_t and monotonicity breaks. That is, the validity range of the candidate formula is at and below the electroweak scale, and the sign flip of the difference and the non-monotonicity appear together upon entering the electroweak range. At the pole the departure vanishes. This is the quantitative form of the theorem: on the domain-time tier Q is exactly $2/3$, and the moment the tier is left, one Schwinger share is added. As an algebraic corollary the amplitude departure also closes: $r^2 = 6Q - 2$ gives $k_L^2(M_Z) = r^2/2 = 1 + 3\,\alpha(M_Z)/(2\pi)$ (Zenczykowski notation $k_L = r/\sqrt{2}$), where 3 is the number of CAS steps (measured 1.003774 vs 1.003732).

Stage-measure discrimination. Which measure of Proposition 1 stands in the denominator of the candidate formula is discriminated by measurement.

Table 11. Stage-measure discrimination - against the measured departure $dQ(M_Z) = +1.258 \times 10^{-3}$.

Candidate measure	**Value**	**Diff.**	**Verdict**
$\alpha/(2\pi)$ - reversible full sphere (Prop. 1)	1.2441×10^{-3}	−1.1%	adopted
α/π - irreversible hemisphere	2.4882×10^{-3}	+98%	excluded
$\alpha/(4\pi)$	6.2204×10^{-4}	−51%	excluded
$\alpha/3$ - CAS steps	2.6056×10^{-3}	+107%	excluded
α^2 order	6.1102×10^{-5}	−95%	excluded

Only the reversible full sphere 2π that Proposition 1 assigns to the reversible axes is adopted; every other candidate, including the irreversible hemisphere π, is excluded. The measure distinction of the axioms is thereby discriminated by measurement. The single-share property is also supported by measurement: the share count $dQ \times 2\pi/\alpha(\mu)$ stays between 0.992 and 1.014 from m_c to m_t.

Antisymmetry of the phase-angle departure. From the same input, the phase-angle departure is $\delta_L(M_Z) - 2/9 = -1.180 \times 10^{-3}$, equal in size and opposite in sign to the Q departure (ratio -0.9377 ± 0.002). This ratio is distinguished from −1 (the full share reversed) by 6.7%, and matches the registered-number ratio $-15/16 = -0.9375$ to 0.02% (hypothesis, card H-994). It is recorded as a structural candidate in which the phase and Q split one share in opposite directions. Zenczykowski describes the M_Z departures as about 0.2% for k_L and about 0.5% for δ_L [5]; the present computation gives k_L +0.19% and δ_L −0.53%, both agreeing with that description.

Remaining tasks. The candidate carries Class III. The remaining tasks are four: formalizing the uniform allocation and the sign of the share, a two-loop refinement (including W contributions), a formal account of the sign flip and the non-monotonicity at the electroweak threshold, and a derivation of the phase-ratio hypothesis $-15/16$ (card H-994). Once the formal derivation is completed, the class of Section 4 is promoted to I. The discrimination window is given in Section 8.3.4.

5. Identification of the Amplitude $r^2 = 2$

Proposition 2 (Identification of the amplitude). $r^2 = 2$ *is the two reversible axes (observer, superposition) of the cost descriptive degrees of freedom. Class III.*

In the cost accumulation partition $(5, 2)$ there are five irreversible axes and two reversible axes. The reversible axes carry no order, hence cost 0, and this 2 is the registered cost degree of freedom.

Non-uniqueness of the identification. The identification is not unique. At least two alternatives exist within the complete system: one is that the norm of the CAS state 011 (Read and Compare active) is $\sqrt{2}$; the other is that the unit-circle norm of the two orthogonal brackets of the Banya equation is $\sqrt{2}$. Card D-09 records the latter. The three identifications give the same number, and the present paper has picked one of them. The class is therefore III, not a derivation. The prior literature places the weak-basis interpretation at this position (Section 7.1); the present paper does not adjudicate between its identification and the weak-basis interpretation.

5.1. Numerical Reproduction (a Prior-Work Confirmation)

The numbers of this subsection are not new findings but recomputation checks of facts confirmed by prior work. The overall constant μ is fixed by the mean of the three root masses. The measured tau enters the constant, so the table is a check, not a prediction.

Table 12. Pole-mass reconstruction - $r^2 = 2$, $\theta = 2/9$, constant fixed by the mean of the three root masses.

k	**Reconstructed (MeV)**	**Measured (MeV)**	**Deviation**
1 (electron)	0.510985	0.51099895	0.00264%
2 (muon)	105.656621	105.6583755	0.00166%
0 (tau)	1776.937991	1776.93	0.00045%

Label assignment. (e, μ, τ) is assigned $k = (1, 2, 0)$. Of the six permutations only this assignment works.

6. Identification of the Phase Angle $\theta = 2/9$

Proposition 3 (Identification of the phase angle). $\theta = 2_{\textit{reversible axes}}/9_{\textit{structural complete description}}$, *with no π conversion entering. Class III.*

6.1. Why π Does Not Enter

By Proposition 1, norm reading is valid only for irreversible + crossings. Reversible axes carry no order, hence cost 0, and with cost 0 no arc is drawn. The numerator 2 is the two reversible axes, hence no arc and no π. The denominator 9 belongs to the structural bracket and is not subject to the norm reading of cost.

This must not be confused with the stage 2π of Section 3.2. That the circumference of the circle is 2π and that the number marking a position on it carries no π are different layers; see item 2 of the Reader's Guide, Section 2.6.

6.2. The Limit of the Dimensionless Argument

The radian is dimensionless by definition, so the statement "a dimensionless ratio is directly a phase angle" is true of every dimensionless number and excludes no candidate. What actually performs the exclusion in Section 6.3 below is the comparison with measurement. The identification is therefore a fit, not a derivation, and its class is III.

Furthermore, under cyclic relabeling of k the angle θ shifts by $2\pi/3$, so $2/9 + 2\pi/3$ gives the same masses equally well. "No π can enter in principle" is true only under the label assignment of Section 5.1.

6.3. Exclusion of the π Forms

For each candidate we list both the value with the constant μ refitted and the value with the constant fixed at the $2/9$ fit. Reading only the fixed-constant column inflates the deviations by up to a factor of 90.

Table 13. Measured comparison of phase-angle candidates - maximum deviation of the mass reconstruction.

Candidate	θ (rad)	Constant refitted	Constant fixed
$2/9$	0.222222	0.0015%	0.0026%
$2/(9\pi)$	0.070736	96.8%	2560%
$2\pi/9$	0.698132	97.7%	6545%
$2/(2+9\pi)$	0.066063	97.1%	2702%
$\pi/9$	0.349066	63.3%	327%

The optimal phase back-solved from measurement is 0.2222248, at relative difference 1.14×10^{-5} from $2/9$.

6.4. The Status of φ_0

Q and the mass ratios are invariant under a global rotation by the $\mathbb{Z}_3$ identity. $\theta = 2/9$ is a specific choice of the global rotation and enters only the reconstruction of individual masses. The Koide ratio $Q = 2/3$ itself is determined by r alone; θ does not enter.

7. Relation to Prior Interpretations

7.1. The Weak-Basis Interpretation

Zenczykowski adopts the proposal that reads $k_f = 1$ as a property of the weak basis; quoting the original, "we accept the suggestion of ref. [12] that $k_f = 1$ is a feature of the weak basis" [5,6]. The identification of the present paper (Section 5) is an alternative to this weak-basis interpretation. The superiority of either interpretation is outside the scope of the present paper (Class IV).

7.2. Attribution of the Values

Zenczykowski attributes the observation that the extracted value 0.2222324 is very close to $\delta_L = 2/9$ "as observed by Brannen and Rosen" [5]. That is, the observation of the values is due to Brannen and Rosen [3,4], and the contribution of the present paper is not the values but the identifications and the scale-tier theorem.

7.3. Caution on Number Overlap: Quark Phases

The prior literature presents $\delta_D = 4/27$ as the phase angle of the down-quark sector [5]. The number overlaps with card D-22 of this series, which uses $\sin\theta_{13} = 4/27$, but the physical quantities differ. The literature values are $\delta_U = 2/27 = \delta_L/3$ and $\delta_D = 4/27 = 2\delta_L/3$, which close in registered numbers as $\delta_U = (2/9)/3$ and $\delta_D = (2/9)(2/3)$. The possibility that the overlap is not accidental but two appearances of the same registered-number combination is recorded as a hypothesis (card H-993); the structural assignment is incomplete.

8. Auxiliary Results

8.1. The Contraction Overlap Ratio Rule (Class III)

Several cards of this series use the form $f = 1 - \ell/N$. The rule that the present paper codifies is the following. N is the data type and is decided by the object. The numerator is a registered descriptive degree of freedom. $\ell = N-$ numerator is determined automatically. Figure 3 summarizes the decider of each of the three slots.

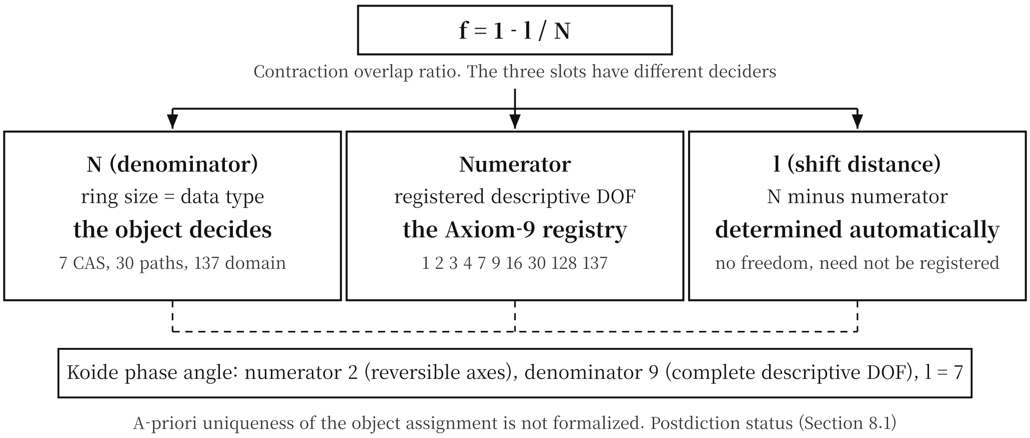

Figure 3. The three slots of the contraction overlap ratio $f = 1 - \ell/N$ and their deciders - the denominator is decided by the object, the numerator by the number registry, and the shift distance is determined automatically.

Three layers of this rule. First, the rule inverts the causality of the complete-system original: in the original, ℓ is the physical distance between two juims and the numerator is the derived value; the present paper restricts the numerator to registered numbers and makes ℓ the derived value. This is a new rule of the present paper. Second, defining $\ell = N-$ numerator makes $f =$ numerator$/N$ hold identically, so the rule itself has no content; the freedom has merely moved from ℓ to the numerator. Third, whether the object assignment that decides N is uniquely determined before seeing the measurement has not been formalized. The rule therefore carries postdiction status. The quantitative chance-hit probability is given in Section 9.1.

8.2. The Neutrino Sector (Class III)

8.2.1. Compare Omission and the Amplitude

The neutrino extension of the Koide relation was examined by Rodejohann and Zhang [9]. The extension of this subsection differs: it changes the amplitude itself into a registered number. Neutrinos omit Compare, fail to acquire the lock, and remain in the OPERATOR bracket; charged leptons cross into the DATA bracket via Compare true. The form follows card P-638:

$$r_\nu^2 = 2\left(1 - \frac{1}{4}\right) = \frac{3}{2}, \tag{5}$$

where 2 is the charged-lepton amplitude and $1/4$ is the share of one domain out of four. The notation "CAS steps 3 divided by brackets 2" would mean occupying 3 slots on a ring of size 2 and does not hold.

8.2.2. Prediction

$\sqrt{m_1} = 0$ forces $\cos\varphi_1 = -1/r_\nu$ and eliminates the phase freedom; the ratio of the two remaining masses is then determined by r_ν alone. Written in the Koide form, the claim of this subsection concerns a single observable:

$$Q_\nu = \frac{3 + \frac{3}{2}r_\nu^2}{9} = \frac{7}{12} = 0.583333. \tag{6}$$

Table 14. Predictions of the two neutrino-amplitude candidates.

r_ν^2	**Predicted** m_3/m_2	**Vs. measurement**
2 (charged-lepton value)	13.928	fully excluded
3/2	5.6634	1.67σ

Data source. The measured $m_3/m_2 = 5.7924 \pm 0.0774$ is the NuFIT 6.0 value (without SK atmospheric data) [11]. Results for both releases are listed.

Table 15. Measured comparison by oscillation-data release.

Data combination	m_3/m_2	**Vs. prediction 5.6634**
NuFIT 6.0, without SK	5.7924 ± 0.0774	1.67σ
NuFIT 5.3, without SK	5.8212 ± 0.0882	1.79σ

8.2.3. Sensitivity to the $m_1 = 0$ Premise

The prediction depends entirely on this premise. Leaving m_1 free dissolves the predictive power.

Table 16. Sensitivity to the m_1 premise.

m_1 **(eV)**	**Measured** m_3/m_2	**Vs. prediction 5.6634**
0	5.7924	$+1.67\sigma$
0.0010	5.7552	$+1.19\sigma$
0.0019	5.6617	-0.02σ
0.0030	5.4827	-2.34σ

$m_1 = 0.0019$ eV is at the level of one tenth of the m_1 allowed by the cosmological sum bound. The figure 1.67σ is therefore meaningful only under the unverified premise $m_1 = 0$.

8.2.4. Opposing Record in the Prior Literature

Quoting Zenczykowski, it is known that the original Koide formula does not work when the charged-lepton masses are replaced with those of neutrinos or quarks, and for neutrinos one estimates directly from experiment that $k_\nu \leq 0.81$ [5]. The $r_\nu^2 = 3/2$ of the present paper is $k_\nu = 0.866$ in that notation and exceeds the quoted bound. The approximation used by the source of that bound, however, is 0.8% high even when applied to the charged leptons, so it is not an immediate ground for rejection. The neutrino amplitude is Class III and depends on the $m_1 = 0$ premise.

8.2.5. Conditions for Candidate Exclusion

The condition for sole survival. The sole survival of $3/2$ holds only when the π forms are excluded from the candidates. Including the π forms, four candidates fit better than $3/2$.

Table 17. Candidate comparison including the π forms.

Candidate	**Value**	**Predicted** m_3/m_2	**Discrepancy**
$(1+4\pi)/9$	1.50737	5.7457	0.60σ
$(4+5\pi)/13$	1.51600	5.8429	0.65σ
$(7+4\pi)/13$	1.50511	5.7203	0.93σ
$15/\pi^2$	1.51982	5.8864	1.21σ
$3/2$ (this paper)	1.50000	5.6634	1.67σ
$13/9$	1.44444	5.0693	9.34σ

The precise statement is the following. Under the π-exclusion rule, $3/2$ survives alone among the pure-quotient candidates. The ground of the π-exclusion rule is that neutrinos do not acquire the lock, are not irreversible, and hence draw no arc. The timing problem remains, however, that the rule was applied after the better-fitting candidates had been seen.

8.3. The Koide Deviation and the Tau Mass

8.3.1. Definition

The defining equation of the deviation is

$$\delta K \;=\; Q - \frac{2}{3} \;=\; -C\,\alpha^3. \tag{7}$$

The exponent 3 of α^3 means one correction at each of the CAS 3 steps. The structure of the coefficient C is the access paths divided by the brackets, $30/2$; the canonical form applying the relation of Section 8.1 is $(2+9\pi)/2 = 15.137$.

8.3.2. Indiscriminable

This subsection is a prediction, not a claim. The sensitivity is $dQ/dm_\tau = 5.646 \times 10^{-5}\ \mathrm{MeV}^{-1}$: 1 keV of tau mass moves the deviation by 5.6×10^{-8}.

Table 18. Deviation and required coefficient as functions of the tau mass.

m_τ **(MeV)**	$Q - 2/3$	**Required** C	**Remark**
1776.84	-7.284×10^{-6}	18.745	PDG 2024 -1σ
1776.86	-6.155×10^{-6}	15.840	earlier-edition central value
1776.93	-2.203×10^{-6}	5.670	PDG 2024 central value
1777.02	$+2.878 \times 10^{-6}$	-7.405	PDG 2024 $+1\sigma$, sign flip

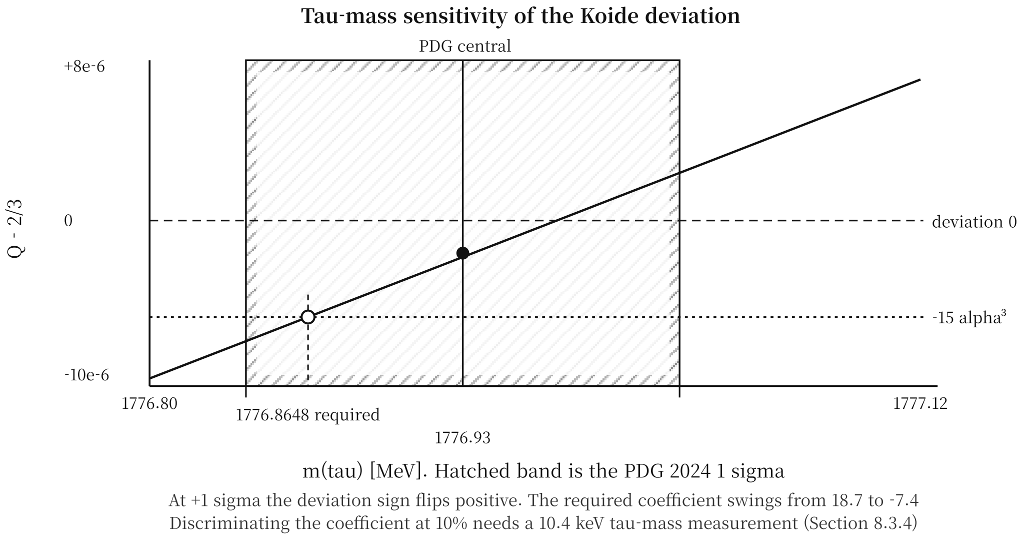

Figure 4. Tau-mass sensitivity of the Koide deviation - within the PDG 2024 1σ band the value and the sign of the deviation pass through the indiscriminable range.

8.3.3. Current Data Prefer No Deviation

The support for the two hypotheses is asymmetric.

Table 19. Required tau mass of the two hypotheses.

Hypothesis	Required m_τ (MeV)	From PDG 2024 central
no deviation (Q exactly $2/3$)	1776.9690	$+0.43\sigma$
this paper's coefficient $(2+9\pi)/2$	1776.8648	-0.72σ

That is, current data support the no-deviation hypothesis better than the coefficient of the present paper. The deviation coefficient is Class IV, indiscriminable in both value and sign at current precision.

8.3.4. Falsification Conditions

Table 20. Falsification conditions and discrimination window of Section 8.3.

Condition	Consequence
m_τ settles near 1776.87 MeV	the coefficient $(2+9\pi)/2$ is supported
m_τ settles near 1776.97 MeV	no deviation is confirmed; this subsection is discarded
m_τ settles at $\geq$ 1777.0 MeV	the deviation sign flips to positive
discriminating the coefficient at 10% precision	requires a 10.42 keV tau-mass measurement, about a 9-fold improvement on the current 90 keV
discriminating between the no-deviation hypothesis and the $(2+9\pi)/2$ coefficient	the required tau masses are 104.2 keV apart; 3σ discrimination needs 34.7 keV, 5σ needs 20.8 keV

8.3.5. The Tau Mass Becomes a Prediction

If $r^2 = 2$ and $\theta = 2/9$ are fixed values, the only free parameter is the overall constant; fixing it by the electron mass alone, the tau mass becomes a prediction. Fixing the constant by the muon or by combinations moves the prediction between 1776.968 and 1776.985 MeV (0.42 to 0.61σ).

Table 21. Tau-mass prediction.

Quantity	Value
predicted m_τ	1776.985 MeV
measured m_τ	1776.93 ± 0.09 MeV
difference	55 keV, 0.61σ

This is the single quantitative condition aimed simultaneously at both identifications of Sections 5 and 6, and it is decided by the same measurement as the deviation adjudication of Section 8.3.4.

9. Discussion

9.1. Restriction of the Search Space: Chance-Hit Probability and Multiple-Comparison Correction

The lattice-density probability is computed as

$$p \;=\; 2 \times (\text{relative error}) \times \sum_j \frac{a_j}{N}, \qquad a_j \in \text{structural registered numbers},\; a_j < N. \tag{8}$$

Table 22. Lattice-density probability - three cases of this series.

Case	N	Numerator candidates	Error	p
$\sin^2\theta_{23}$	7	1, 2, 3, 4	0.10%	0.0029
$\sin^2\theta_W$	30	1, 2, 3, 4, 7, 9, 16	0.91%	0.0255
$\sin^2\theta_{13}$	137	1, 2, 3, 4, 7, 9, 16, 30, 128	0.46%	0.0134

Multiple-comparison correction. Reading the product of the three values, 9.9×10^{-7}, as a significance level does not hold: the mining cards of this series number 1794, and the product is over 3 well-fitting cases selected from them. Correcting for the number of trials:

$$p_{\text{global}} = 1 - (1 - 9.9 \times 10^{-7})^{1794} = 1.77 \times 10^{-3} \quad (\text{about } 2.9\sigma)..$$

This corrected value is itself an upper bound before the combination forms below are taken into account. Moreover, $\sum a_j/N$ lies between 1.40 and 1.46 in all three cases, so p is effectively proportional to the relative error alone; the candidate-count column does not affect the result. Taking the combination forms (simple quotients, overlap ratios, π forms, orthogonal sums $a/(b + c\pi)$) broadly widens the room for a number of 1% precision to land on the lattice. This is why the present paper assigns the two parameter identifications to Class III (postdiction) and does not present them as derivations.

9.2. Comparison with Prior Approaches

Table 23. Comparison of prior approaches to grounding the Koide relation.

Approach	Method	Target	Difference from this paper
Foot [7]	geometric interpretation: $Q = 2/3$ is a condition on the angle between the root-mass vector and $(1, 1, 1)$	Q itself	a re-expression; does not treat the parameter values or the scale behavior
Sumino [8]	family gauge symmetry effective theory; QED-correction cancellation mechanism	stability of the relation	introduces new gauge fields; this paper assigns the scale by tier separation without new fields
Koide yukawaon [2]	scalar vacuum-expectation-value model	$Q = 2/3$	introduces new scalars; no identification of the parameter values
Gerard, Goffinet, Herquet [6]	weak-basis interpretation: $k_L = 1$ is a property of the weak basis	$k_L = 1$	the competing alternative to Section 5; superiority not adjudicated
this paper	registered-number identification + time-tier separation	r^2, θ, validity scale	derives the scale tier as a prediction (Section 4); the value identifications are Class III

9.3. Scope and Limits

Table 24. Scope and limits - organized by class.

Item	Class	Content
$\mathbb{Z}_3$ parametrization	II	a discrete-Fourier-transform identity; no physical input
stage 2π	III	application of Proposition 1: reversible means full sphere
scale tier (pole mass)	III	center of this paper; direction predicted (theorem), size a candidate $\alpha(\mu)/(2\pi)$ (Section 4.6)
amplitude $r^2 = 2$ identification	III	three alternative identifications exist; not unique
phase $\theta = 2/9$ identification	III	the exclusion is performed by measurement; not a derivation
contraction overlap ratio rule	III	postdiction; the rule itself is identically empty
neutrino amplitude	III	entirely dependent on the $m_1 = 0$ premise
deviation coefficient	IV	indiscriminable in value and sign; current data prefer no deviation

The remaining limits are the following.

- **Non-uniqueness of the identifications.** The same number 2 is registered in both the structural and the cost brackets, and the source cards read it differently. The choice of this paper is one reading, not a derivation.
- **The departure size is at the candidate stage.** Section 4.6 presents the size candidate $\alpha(\mu)/(2\pi)$; the formal derivation of the Schwinger share and the comparison against precision $\alpha(\mu)$ tables remain. Upon completion, the class of Section 4 is promoted to I.
- **Quantification of the deviation.** Not decidable before the tau-mass precision improves to the 10 keV level; current data favor no deviation.
- $m_1 = 0$**.** A premise, not a derivation. A rise of only 0.0019 eV dissolves the prediction of Section 8.2.
- **Superiority against the weak-basis interpretation.** The present paper does not adjudicate which of the prior interpretation and its identification is correct.

9.4. Classification of All Results

Table 25. All results of the present paper - on the 4 axioms + 1 proposition.

Class	Result	Remark
II	equivalence of $Q = 2/3$ and $r^2 = 2$	$\mathbb{Z}_3$ discrete-Fourier identity; no physical input (Section 3.1)
III	ground of the stage 2π and the 120-degree spacing	application of the reversible full sphere of Proposition 1 (Section 3.2)
III	scale-tier theorem: holds at the pole masses	juim recording into DATA + Paper I correspondence rule; agrees with Zenczykowski's observation [5]; center of this paper (Section 4)
III	departure-size candidate $Q(\mu) - 2/3 = \alpha(\mu)/(2\pi)$	agreement 0.2 to 0.8% below the electroweak range and within 1.4% in it; validity range at and below the electroweak scale; the stage-measure discrimination adopts 2π only; formal derivation subsequent (Section 4.6)
III	$k_L^2(M_Z) = 1 + 3\alpha(M_Z)/(2\pi)$	algebraic corollary; measured 1.003774 vs 1.003732 (Section 4.6)
III	amplitude $r^2 = 2$ identification	two reversible axes; alternatives exist (Section 5)
III	phase $\theta = 2/9$ identification	reversible 2 over structural 9; exclusion performed by measurement (Section 6)
III	codification of the contraction overlap ratio rule	postdiction status (Section 8.1)
III	neutrino $r_\nu^2 = 3/2$, $m_3/m_2 = 5.6634$	dependent on the $m_1 = 0$ premise; 1.67σ (Section 8.2)
III	tau-mass prediction 1776.985 MeV	the quantitative test of both identifications; 0.61σ (Section 8.3.5)
IV	deviation coefficient $(2 + 9\pi)/2$	indiscriminable; current data prefer no deviation (Section 8.3)
IV	superiority against the weak-basis interpretation	not adjudicated (Section 7.1)

Classification criteria - I: formally derived from the axioms. II: the axioms force a structure and mathematics fixes it automatically. III: axiom-described in full - independent formal proof is subsequent. IV: not included in the scope of the present paper. The criteria are identical to Paper I [13].

10. Conclusion

The $\mathbb{Z}_3$ parametrization of the Koide relation and its parameter values are already registered in the prior literature, and prior interpretations exist. The present paper has answered the three questions of the Introduction: Q1 and Q2 by registered-number identifications (Sections 5 and 6, Class III), and Q3 by the scale-tier theorem (Section 4).

The causal chain of the derivation is: Axiom 1 (two time tiers) ==> juim recorded into DATA (Axiom 2) ==> mass is a domain-time-tier quantity ==> threshold-included definition ==> holds at the pole masses, departs at high scales. The candidate for the departure size is the Schwinger share $\alpha(\mu)/(2\pi)$, which agrees across the scales (Section 4.6).

Table 26. Summary of the results of the present paper.

Result	Content	Class
scale tier	mass is recorded into DATA, hence a domain-time-tier quantity, hence the relation holds at the pole masses; forms the opposite end to Paper I's system-time case, and the departure-size candidate $\alpha(\mu)/(2\pi)$ agrees across the scales (Section 4.6)	III
amplitude	$r^2 = 2$ is the two reversible axes; alternatives exist	III
phase angle	$\theta = 2/9$ is reversible 2 over structural 9; reversible, hence no arc	III
tau mass	predicted 1776.985 MeV, 0.61σ from measurement	III

The unique predictions of the present paper are four, each with a concrete falsification condition.

Table 27. Unique predictions of the present paper - deductive consequences of the axiom structure.

#	Prediction (axiomatic ground)	Source	Falsification condition
1	**Scale tier.** The Koide relation holds at the pole masses and breaks at threshold-free scales; juim recording into DATA + time-tier separation	Sec. 4	holds at the M_Z scale with pole-level precision
2	**Departure size.** $Q(\mu) - 2/3 = \alpha(\mu)/(2\pi)$, with corollary $k_L^2(M_Z) = 1 + 3\alpha(M_Z)/(2\pi)$; the share of one coupling over the reversible circle 2π	Sec. 4.6	systematic disagreement against precision mass tables
3	**Tau mass 1776.985 MeV.** The two identifications + the constant fixed by electron and muon	Sec. 8.3.5	tau-mass precision settles 3σ away from 1776.985; window in Table 20
4	**Neutrino $m_3/m_2 = 5.6634$.** Compare omission + $r_\nu^2 = 3/2$ (premise $m_1 = 0$)	Sec. 8.2	oscillation data settle 3σ away, or $m_1 > 0.002$ eV is established

Table 28 compares the scope of derivations of the present paper (4 axioms + 1 proposition) with the complete axiomatic system [16] (15 axioms). The present paper treats only the minimal axiom set needed for the Koide relation.

Table 28. Scope of derivations - this paper versus the complete system.

Item	This paper	Complete system [16]
Number of axioms	$4 + 1$ prop.	15
Derivation items (Table 25 classification)		
Experimentally matched ($< 1\%$)	3	155
Structural/mathematical results	6	-
Awaiting verification	3	941
Subtotal	12	1,096
Unique predictions (experimentally testable)		
Awaiting verification	4	130

Paper I obtained the coupling constant from the signature $(5,2)$, Paper II the metric of gravitational quantization from the same $(5,2)$, and Paper III the structural constants of black hole thermodynamics. The present paper shows that the reversible-axis 2 of the same $(5,2)$ appears in both the amplitude and the phase angle of the generation structure, and further confirms, as a second case, that the separation of system time and domain time assigns to each physical quantity the scale at which its relation holds.

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Conflicts of Interest: The author declares no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CAS	conditional state transition (Compare-And-Swap)
R, C, S	observe (Read), compare (Compare), record (Swap)
DOF	degrees of freedom
QED	quantum electrodynamics
PDG	Particle Data Group
MS	minimal subtraction scheme

Appendix A. Numerical Evaluation

PDG 2024 basis [10]: $\alpha = 1/137.035999177$, $m_e = 0.51099895000$ MeV, $m_\mu = 105.6583755$ MeV, $m_\tau = 1776.93 \pm 0.09$ MeV. Neutrinos: NuFIT 6.0 without SK [11], $\Delta m^2_{21} = (7.49 \pm 0.19) \times 10^{-5}$ eV^2, $\Delta m^2_{31} = (2.513 \pm 0.021) \times 10^{-3}$ eV^2.

Table A1. All numbers of the present paper.

Quantity	Value
measured Q	0.666664463 ± 0.000005
departure of Q from $2/3$	0.43σ
$Q - 2/3$	-2.203×10^{-6}
dQ/dm_τ	5.646×10^{-5} MeV^{-1}
$2 + 9\pi$	30.27433
$(2 + 9\pi)/2$	15.137
measured optimal phase	0.2222248, 1.14×10^{-5} from $2/9$
charged-lepton masses at M_Z (XZZ [12])	0.486570154, 102.7181337, 1746.17 ± 0.15 MeV
measured $Q(M_Z)$	0.667924713
$k_L^2(M_Z)$, $\theta(M_Z)$	1.003774, 0.2210426
$\delta_L(M_Z) - 2/9$	-1.180×10^{-3} (ratio $d\theta/dQ = -0.9377$)
$1/\hat{\alpha}(m_\tau)$ anchor (PDG 2024 [10])	133.450
$Q(M_Z) - 2/3$	$+1.258 \times 10^{-3}$
$\alpha(M_Z)/(2\pi)$	1.2441×10^{-3} ($1/\hat{\alpha}(M_Z) = 127.930$)
$1/(256\pi)$	1.2434×10^{-3}
discrimination gap of the hypotheses (Section 8.3.4)	104.2 keV (10.42 keV for 10% discrimination)
measured m_3/m_2	5.7924 ± 0.0774
predicted m_3/m_2	5.6634
predicted tau mass	1776.985 MeV

Constant-fixing method. The reconstruction of Section 5.1 fixes the overall constant by the mean of the three root masses; the tau prediction of Section 8.3.5 fixes it by the electron mass alone. The exclusion table of Section 6.3 lists both the refitted-constant and the fixed-constant values per candidate.

Appendix B. Registry Cross-Table

Table A2. Bracket identity of the registered numbers appearing in this paper.

No.	Bracket	Identity	Used in this paper as
2	cost	two order-free axes	$r^2 = 2$, numerator of θ
2	structure	two brackets	denominator of r_ν^2, denominator of the deviation coefficient
9	structure	structural complete descriptive DOF $7 + 2$	denominator of θ
9	cost	residual cost $13 - 4$	the 9π in the denominator of $\sin^2\theta_W$
30	structure	number of access paths	numerator of the deviation coefficient
15	-	not registered; holds only as the quotient $30/2$	deviation coefficient

Same number, different brackets. The numbers 2 and 9 are registered in both the structural and the cost brackets with different identities. This paper fixes, per position, which one is meant, by the table above. The very existence of this freedom is what pins the identifications at Class III.

Appendix C. Exhaustive Exclusion Tables

The tables of this appendix are exhaustive within the stated candidate spaces, and the candidate spaces themselves depend on rule choices.

Appendix C.1. Phase-Angle Candidates

Identical to Table 13 of Section 6.3.

Appendix C.2. Neutrino-Amplitude Candidates

All candidates within 3.5σ across the combination forms used by this series are listed. Including the π forms, $3/2$ is not the best fit.

Table A3. Exhaustive neutrino-amplitude candidates - before and after the π-exclusion rule.

Candidate	Value	m_3/m_2	Discrepancy	Under π exclusion
$(1+4\pi)/9$	1.50737	5.7457	0.60σ	dropped
$(4+5\pi)/13$	1.51600	5.8429	0.65σ	dropped
$(7+4\pi)/13$	1.50511	5.7203	0.93σ	dropped
$15/\pi^2$	1.51982	5.8864	1.21σ	dropped
$3/2$	1.50000	5.6634	1.67σ	survives
$13/9$	1.44444	5.0693	9.34σ	survives, excluded
$7/5$	1.40000	4.6240	15.10σ	survives, excluded
$5/3$	1.66667	7.7485	25.27σ	survives, excluded

Appendix D. Prior Records

Table A4. Prior records in the card library of this series and their correspondence in this paper.

Card	Content	Correspondence
D-45	$2/9 = (1-7/9)$ contraction-overlap-ratio structure; numerator 2 recorded as the bracket count	Sec. 6, 8.1
D-09	parameters $\theta = 2/9$, $r = \sqrt{2}$; r recorded as the unit-circle norm	Sec. 5, 6
H-22	the numerator of $2/9$ recorded as the Compare degree of freedom	Sec. 6
D-14, D-27	Koide deviation and coefficient decomposition	Sec. 8.3
D-56	$\sin^2\theta_W = 7/(2+9\pi)$	Sec. 4, 8.1
H-05	neutrino Compare omission	Sec. 8.2
D-05	$\sin^2\theta_{12} = 3/\pi^2$; a geometric ratio internal to the OPERATOR bracket	Sec. 8.2
P-638	neutrino amplitude $r_\nu^2 = 2(1-1/4) = 3/2$	Sec. 8.2
H-992	Koide scale departure $Q(\mu) - 2/3 = \alpha(\mu)/(2\pi)$	Sec. 4.6
H-994	phase-departure ratio $d\theta/dQ = -15/16$, hypothesis	Sec. 4.6
H-993	quark-phase registry decomposition $\delta_U = (2/9)/3$, $\delta_D = (2/9)(2/3)$	Sec. 7.3

Non-uniqueness of the identification. The numerator 2 of $\theta = 2/9$ is read by D-45 as the bracket count, by H-22 as the Compare degree of freedom, and by this paper as the reversible axes. The three readings give the same number, and this non-uniqueness is the ground for placing the identification at Class III.

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